

Dinosaurs A large group of archosaurs with upright limbs. Dinosaurs were the dominant land animals for 160 million years.

DNA Deoxyribonucleic acid, a chemical whose molecules carry genetic instructions from one generation to the next in nearly all organisms. (See Genes.) The complex double-helix structure of DNA was discovered in the 1950s.

Domesticated Bred to be tame. Cows, sheep, and dogs are examples of domestic animals.

Dromaeosaurs A group of bird-like, two-legged, carnivorous dinosaurs. Most grew no longer than 2 m (6 ft). Dromaeosaurs lived in all northern continents.

Echinoderms Marine invertebrates with a hard, chalky skeleton and a five-rayed symmetry. They evolved during the Cambrian Period and include starfish, sea lilies, sea cucumbers, and sea urchins.

Ediacaran biota Fossil organisms named after those found in the Ediacaran Hills of Australia. The sea-dwelling, soft-bodied organisms lived about 550 million years ago and were some of the earliest animals.

Embryo An animal or plant in an early stage of development from an egg or a seed.

Environment The natural surroundings in which an animal or plant lives.

Era A very long unit of time. Eras are divided into shorter units called periods. The Mesozoic Era, for example, is divided into the Triassic, Jurassic, and Cretaceous periods.

Eurypterids (sea scorpions) An extinct group of large arthropods related to modern scorpions. Eurypterids lived in the sea and freshwater habitats during the Palaeozoic Era. Some grew more than 2 m (6 ft 6 in) long.

Evolution The gradual change of animal or plant species over long periods of time. Evolution is driven mainly by a process called natural selection.

Excavation Digging out and removing fossils or other objects from the ground.

Exoskeleton An external skeleton. Animals such as crabs have an exoskeleton. In contrast, humans have an internal skeleton.

Extinction The dying-out of a plant or animal species. Extinction can happen naturally as a result of competition between species, changes in the environment, or natural disasters (such as an asteroid striking Earth).

Fossil The ancient remains or imprint of a prehistoric organism preserved in rock.

Fossilization The process by which dead organisms turn into fossils. Fossilization often involves replacement of the original organism with rock minerals.

Gastropods A class of invertebrates made up of snails, slugs, and their many aquatic relatives, such as cowries and limpets.

Genes Chemical instructions encoded in the DNA molecule. Genes control the way all organisms grow and develop. They are passed on from parents to their young.

Gondwana A vast prehistoric supercontinent that included South America, Africa, Antarctica, Australia, and India. Gondwana existed from Precambrian times until the Jurassic Period, when these lands began to move apart.

Habitat The place where a plant or animal lives naturally.

Hadrosaurids (duck-billed dinosaurs) A family of plant-eating dinosaur species with duck-like beaks that lived during the Cretaceous Period.

Hagfish (Agnathan) A kind of living jawless fish.

Herbivore An animal that eats plants.

Hominins The family of apes that includes humans and our closest relatives.

Homo erectus A prehistoric member of the human family that lived from 2 million years ago to 100,000 years ago. *Homo erectus* evolved in Africa but spread to Asia.

Homo sapiens The scientific name for modern human (our own species).

Hybrid The offspring of parents from two different species.

Ichthyosaurs Prehistoric marine reptiles that resembled dolphins or fish.

Iguanodontians Large, plant-eating ornithomimid dinosaurs that were common in the Early Cretaceous.

Invertebrates Animals without backbones.

Jawless fish (Agnathans) A class of primitive vertebrates that flourished mainly in Early Palaeozoic times. They include extinct groups and the living hagfish and lampreys.

Jurassic The middle period in the Mesozoic Era, from 200 to 145 million years ago. During the Jurassic Period, dinosaurs dominated the land, the first birds evolved, and mammals began to diversify.

Juvenile A young animal.

Kin Family – individuals that are genetically related.

Lamprey A type of living jawless fish with a round, sucker-like mouth.

Lungfish A type of fish that has both gills and lungs and can breathe in water and air. Lungfish appeared in the Devonian Period.

Mammals Warm-blooded, hairy vertebrates that secrete milk and suckle their young. Living mammals range from tiny shrews to the blue whale (the largest creature ever) and occupy a great variety of habitats. Mammals originated in the Triassic Period.

Marine Belonging to the sea (particularly animals or plants).

Marsupial A type of mammal with a pouch in which the young develop. Living examples include kangaroos and wallabies. Marsupials survive only in Australasia and the Americas.

